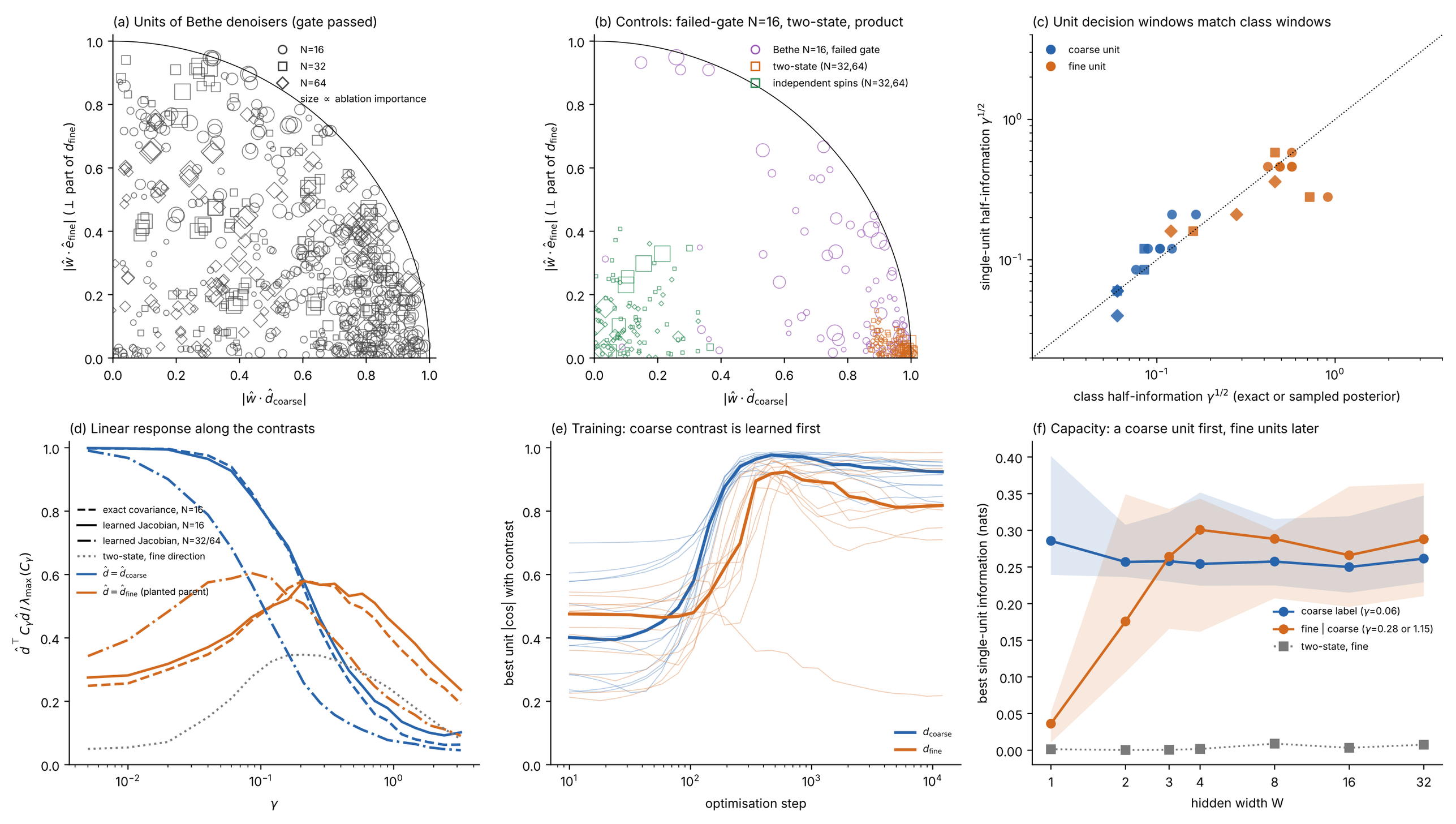

(a) Units of Bethe denoisers (gate passed)
N=16
N=32
N=64
size ∝ ablation importance
$|\hat{w}\cdot\hat{e}_{\text{fine}}|$ (⊥ part of d_{fine})
$|\hat{w}\cdot\hat{d}_{\text{coarse}}|$
(b) Controls: failed-gate N=16, two-state, product
Bethe N=16, failed gate
two-state (N=32,64)
independent spins (N=32,64)
$|\hat{w}\cdot\hat{e}_{\text{fine}}|$ (⊥ part of d_{fine})
$|\hat{w}\cdot\hat{d}_{\text{coarse}}|$
(c) Unit decision windows match class windows
coarse unit
fine unit
single-unit half-information $\gamma^{1/2}$
class half-information $\gamma^{1/2}$ (exact or sampled posterior)
(d) Linear response along the contrasts
exact covariance, N=16
learned Jacobian, N=16
learned Jacobian, N=32/64
two-state, fine direction
$\hat{d}=\hat{d}_{\text{coarse}}$
$\hat{d}=\hat{d}_{\text{fine}}$ (planted parent)
$\hat{d}^{\top} C_\gamma \hat{d}/\lambda_{\max}(C_\gamma)$
γ
(e) Training: coarse contrast is learned first
best unit |cos| with contrast
d_{coarse}
d_{fine}
optimisation step
(f) Capacity: a coarse unit first, fine units later
coarse label (γ=0.06)
fine | coarse (γ=0.28 or 1.15)
two-state, fine
best single-unit information (nats)
hidden width W